NIT RAIPUR INTERAL HACKATHON 2025

TEAM OVERFITTERS

- **Problem Statement ID – 70**
- **Problem Statement Title-** **Multilingual Emergency Communication and Misinformation Filter**
- **Theme-** DISASTER MANAGEMENT
- **PS Category- Software**
- **Team ID-**
- **Team Name (Registered on portal)**

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BEACON

Trusted Crisis Information Network

Idea/Solution:

A **trusted**, **multilingual** crisis network that continues working when **the internet does not.**

- ❖ **Multimodal reporting:** Submit voice, text, images, and emergency requests in Indian languages.
- ❖ **AI Brain Router:** Transcribes, translates, verifies evidence, detects duplicates, prioritizes needs, and suggests routes.
- ❖ **Misinformation detection:** Cross-checks claims against official alerts, nearby evidence, and verified regional news source
- ❖ **Multi-layer connectivity:** Seamlessly switches from internet to SMS and Bluetooth/Wi-Fi Direct, ensuring critical alerts reach users even during network outages.
- ❖ **Authority Command Centre:** Provides live incident maps, rumour detection, human verification, resource coordination and correction controls.
- ❖ ❖ **Accessible for Everyone:** Voice navigation, read-aloud alerts, haptic feedback, pictograms, large text, and screen-reader support make Beacon easy for everyone to use.

Problem Resolution:

- ❖ **Prevents Misinformation and Confusion:** Beacon verifies public reports using official alerts, nearby evidence, and trusted news sources while merging duplicates into one clear incident.
- ❖ **Connects Communities Across Languages:** Beacon translates voice, text, and image reports into Indian languages and connects affected people through temporary local groups.

Unique Value Preposition:

❖ **Local Crisis Support Network:** Beacon connects affected people in temporary emergency groups, enabling them to request or provide nearby help, including food, medicine, shelter, and transport.

❖ **Multilingual, Verified Reporting:** Citizens can submit voice, text, or image reports in Indian languages and receive a clear verification status for every report.

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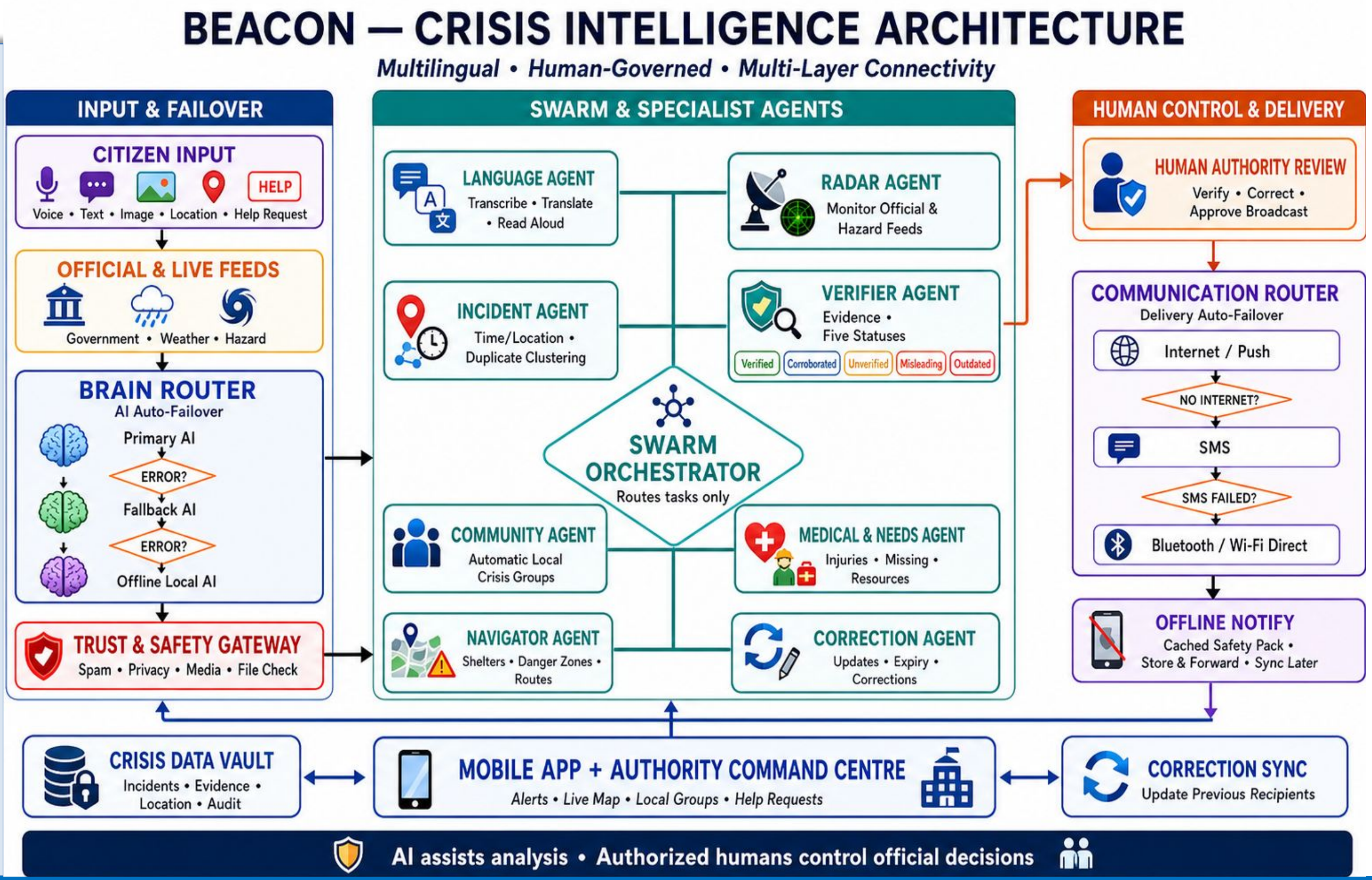
TECHNICAL APPROACH

- **Mobile:** React Native, Expo
- **Frontend:** Next.js, Tailwind CSS
- **Backend:** FastAPI, Python
- **Database:** PostgreSQL, PostGIS, SQLite
- **AI/ML:** Whisper, IndicTrans2, Tesseract, Qwen, Multilingual E5
- **Data Collection:** Scrapy, BeautifulSoup, RSS, Government APIs
- **Maps:** MapLibre, OpenStreetMap
- **Real-Time:** WebSockets, Redis
- **Offline Sharing:** SMS, Bluetooth, Wi-Fi Direct
- **Security:** JWT, AES, Ed25519



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FEASIBILITY AND VIABILITY

Feasibility	Challenges and Risks	Strategies to Overcome Challenges
1. Technical Feasibility • Open-source technology stack • Existing multilingual AI models • Modular and scalable architecture	**1. Information Accuracy** • AI may misclassify reports • Translations may lose context • Old media may appear genuine	**1. Reliable Verification** • Human approval for critical alerts • Rule-based evidence checking • Original content preserved
2. Operational Feasibility • Simple multilingual interface • Authority command dashboard • Offline-first functionality	**2. Data Integration** • Limited government APIs • Unreliable news sources • Different data formats	**2. Trusted Data Integration** • APIs, RSS and permitted scraping • Verified-source allowlist • Standardized data adapters
3. Financial Feasibility • Low-cost open-source tools • Local AI processing • Phased implementation	**3. Connectivity and Scale** • Network failure during disasters • Sudden increase in users • SMS/Bluetooth limitations	**3. Resilient Infrastructure** • Offline caching and report queues • Internet → SMS → Bluetooth/Wi-Fi • Scalable cloud infrastructure
4. Social Feasibility • Indian-language support • Accessible and low-literacy design • Community participation	**4. Privacy and Misuse** • Sensitive location exposure • Fake or duplicate submissions • Misuse of local groups	**4. Security and Moderation** • Encryption and role-based access • Approximate public locations • Rate limits and content moderation

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IMPACT AND BENEFITS

Impact on Target Audience

1. Citizens
- Verified local alerts
- Multilingual reporting
- Clear safety instructions

2. Authorities
- Unified incident dashboard
- Emerging-rumour detection
- Consolidated public requests

3. Responders and Volunteers
- Nearby help requests
- Injury and resource details
- Location-based coordination

4. Vulnerable Communities
- Voice and read-aloud alerts
- Pictograms and large text
- Offline access

Benefits of the Solution

1. Social Impact
- Reduces rumours and panic
- Removes language barriers
- Strengthens community coordination

2. Operational Benefits
- Faster incident assessment
- Reduced duplicate reports
- Better resource allocation

3. Economic Benefits
- Reduces repeated effort
- Uses open-source technologies
- Optimizes available resources

4. Inclusive Communication
- Indian-language support
- Accessible interface
- Multi-layer connectivity

Key Outcomes

1. Trusted Information
- Evidence-based verification
- Clear verification status
- Traceable updates and corrections

2. Faster Response
- Prioritized emergency requests
- Live crisis visibility
- Improved authority coordination

3. Efficient Resource Use
- Faster request assignment
- Reduced response delays
- Data-driven decisions

4. Resilient Communication
- Offline safety guidance
- Internet → SMS → Bluetooth/Wi-Fi
- Continued access during outages

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RESEARCH AND REFERENCES

Details / Links of Reference and Research Work

1. Academic Research Papers

- **CrisisMMD—Multimodal Disaster Analysis:** https://arxiv.org/abs/1805.00713
- **IndicTrans2—Indian-Language Translation:** https://arxiv.org/abs/2305.16307
- **Multilingual E5—Report Matching:** https://arxiv.org/abs/2402.05672
- **Multimodal Crisis-Event Classification:** https://arxiv.org/abs/2004.04917

2. Indian Government Sources

- **NDMA SACHET—Official Disaster Alerts:** https://sachet.ndma.gov.in/
- **IMD—Weather Forecasts and Warnings:** https://mausam.imd.gov.in/
- **NDMA Alerts Hub—Government Bulletins:** https://ndma.gov.in/alert-hub/Alert
- **Data.gov.in—Disaster Datasets:** https://www.data.gov.in/keywords/disaster

3. International Data and Standards

- **USGS—Real-Time Earthquake Data:** https://earthquake.usgs.gov/fdsnws/event/1/
- **GDACS—Global Multi-Hazard Alerts:** https://www.gdacs.org/xml/rss.xml
- **NASA EONET—Natural-Event Tracking:** https://eonet.gsfc.nasa.gov/api/v3/events
- **OASIS CAP 1.2—Emergency-Alert Standard:** https://www.oasis-open.org/standard/cap/